TEXORA · LECTURE SHEET

Sizing Calculations

Sizing is bought by the kilogram and judged at the loom. Its calculations link the recipe in the mixing vessel to the add-on on the yarn and the cost on the fabric: concentration, pick-up, add-on, stretch, and the price of a kilogram of size solids — the chain this sheet works end to end.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 3 · SPECIALIZATION — FABRIC MANUFACTURING

1. The three quantities that must not be confused

- **Concentration** — dry size solids per unit of size liquor, % (kg of solids per 100 kg of mixed liquor).
- **Wet pick-up** — liquor taken up by the warp, as % of the dry unsized warp weight; set by immersion, squeeze pressure and speed.
- **Add-on** — dry size actually remaining on the yarn after drying, as % of dry unsized warp weight. This is the number the loom feels and the specification quotes.

$$\text{Add-on (\%)} = \text{concentration (\%)} \times \text{wet pick-up (\%)} \div 100$$

$$\text{Add-on (\%)} = (\text{sized warp weight} - \text{unsized warp weight}) \div \text{unsized warp weight} \times 100 \quad \text{(dry basis)}$$

WORKED EXAMPLE — ADD-ON FOR A PRODUCTION WARP

A set carries **850 kg** of dry unsized warp and is specified at **11% add-on**.
Dry size required on the yarn = 850 × 0.11 = **93.5 kg of solids**.
At a liquor concentration of 10%, the warp must pick up 93.5 ÷ 0.10 = **935 kg of liquor** — a wet pick-up of 935 ÷ 850 = 110% of the warp weight, which the size-box settings (squeeze pressure lowered, immersion deepened) must actually deliver. If the delivered pick-up is only 95%, the add-on falls to 10 × 95/100 = 9.5% however good the recipe is: add-on is made at the squeeze rollers, not in the mixing vessel.

2. The recipe and the price of a kilogram of solids

INGREDIENT	KG PER BATCH	ROLE	PRICE (TK/KG)	COST (TK)

Modified starch	70	Film former, body	95	6,650
PVA	15	Strength, abrasion resistance	420	6,300
Wax / lubricant	5	Flexibility, surface slip	260	1,300
Softener	2	Film softness	180	360
Total solids	**92**	made up to ≈1,000 L		**14,610**

WORKED EXAMPLE — COSTING THE RECIPE

Batch cost = 6,650 + 6,300 + 1,300 + 360 = **Tk 14,610** for 92 kg of solids.

Cost per kg of solids = 14,610 ÷ 92 = **Tk 158.8 ≈ Tk 159 per kg**.

At 11% add-on, size material cost per kg of warp = 0.11 × 158.8 = **Tk 17.47 per kg of warp**.

Note where the money sits: PVA is 16% of the solids but 43% of the cost. Recipe reformulation is therefore mostly a PVA question — and every point of add-on saved at the same loom performance is worth Tk 1.59 per kg of warp.

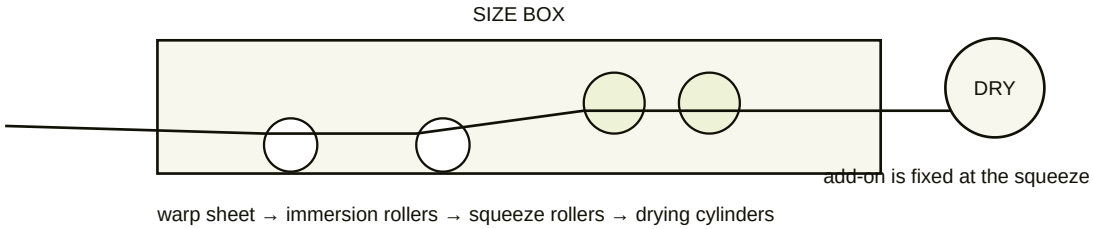


Figure 1 — The size box in section: immersion, squeeze, dry. Concentration is set in the kitchen; add-on is decided at the squeeze rollers.

3. Stretch, contraction and the sized length

The warp leaves the sizing machine longer than it entered: wet yarn under tension stretches, and only part of the stretch recovers. A stretch of 1.2% turns a 1,000 m unsized length into **1,012 m** of sized warp — length the loom plan gains, but bought with yarn tension that reappears as end-breaks if it is pushed further. The planner works the chain both ways: fabric length → grey warp length (add warp crimp, Sheet 5) → sized beam length (the beam must carry the grey length divided by (1 + stretch) of unsized warp, plus loom waste at both ends). Moisture bookkeeping closes the loop: sized warp is traded and stored at a working regain, so 100 kg of oven-dry sized warp at 7% regain weighs **107 kg** on the floor — the difference between the dry-basis add-on calculation and the scale at the sizing creel.

4. Concentration control

On the floor, concentration is read from a refractometer (Brix) or hydrometer against a calibration for the recipe, and corrected by adding water or solids — never by changing squeeze settings to chase a number the kitchen got wrong. A liquor of 25 kg solids made up to 280 kg reads 25 ÷ 280 × 100 = **8.9%**: if the specification is 10%, the batch is short of solids by roughly (0.10 × 280) − 25 = 3 kg, and that, not the rollers, is the fix.

Glossary

Add-on — dry size on the yarn as % of dry unsized warp weight. **Wet pick-up** — liquor taken up, % of dry warp weight. **Concentration** — solids % of the size liquor. **Size solids** — the dry ingredients of a recipe, water excluded. **Stretch** — permanent length gain of the warp during sizing. **Brix** — refractometer scale used to read size concentration on the floor.

Key points

• Add-on = concentration × wet pick-up ÷ 100 (worked: 850 kg warp at 11% needs 93.5 kg solids, 935 kg of 10% liquor).
• Cost the recipe per kg of solids (worked: Tk 14,610/92 kg = Tk 159/kg; 11% add-on = Tk 17.47 per kg warp).
• One ingredient usually dominates cost out of proportion to its mass (PVA: 16% of solids, 43% of cost).
• Stretch adds length (1.2% → 1,012 m from 1,000 m); regain adds scale weight (100 kg dry = 107 kg at 7%).

Practice

1. A sizing programme handles 1,416 kg of warp at 10% add-on of the Section 2 recipe. Find the kg of solids required and their cost.
2. Liquor concentration is 9% and wet pick-up is 105%. What add-on results, and what single size-box change raises it toward 11%?
3. A sized beam must deliver 2,500 m of grey warp length. Sizing stretch is 1.2%. What length of unsized warp must the set carry (ignore loom waste)?
4. Why does the costing use dry solids rather than litres of liquor?

Answer key

1. Solids = 1,416 × 0.10 = **141.6 kg**; cost = 141.6 × 158.8 ≈ **Tk 22,490**.
2. Add-on = 9 × 105 ÷ 100 = **9.45%**. Raise wet pick-up by reducing squeeze-roller pressure (or deepening immersion): pick-up must reach 11 × 100/9 ≈ 122%.

3. Unsized length = 2,500 ÷ 1.012 = **2,470 m**.
4. Because water is (nearly) free and evaporates in the cylinders: two liquors of different concentration deliver the same add-on at different volumes. What the mill buys, stores and pays for is solids; litres only describe the vehicle.
